

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO2: Analyze Machine Learning and Deep Learning models by evaluating neural network architectures, representation learning, activation functions, unsupervised learning techniques, and their real-world applications. (L4)

CO Weightage: 15%

Assessment Tool Weightage: 20%

QUESTIONS:

Total Marks: 20

1. Differentiate between Machine Learning and Deep Learning with respect to architecture, data requirements, feature extraction and applications. Give two real-world applications of each.
2. Explain the concept of Representation Learning. Discuss how the width and depth of a neural network influence its learning capability, computational complexity and model performance.
3. Compare ReLU, Leaky ReLU (LReLU) and ELU (Exponential Linear Unit) activation functions. Explain their mathematical behavior, advantages, disadvantages and suitable use cases.
4. Explain the concept of unsupervised learning in neural networks. Describe the working principle of Restricted Boltzmann Machines (RBMs) and Autoencoders and compare their objectives and applications.

Code of Conduct:

I G J D SAI SHARAN/192425152(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G. J. D. Saq Sharan

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

①

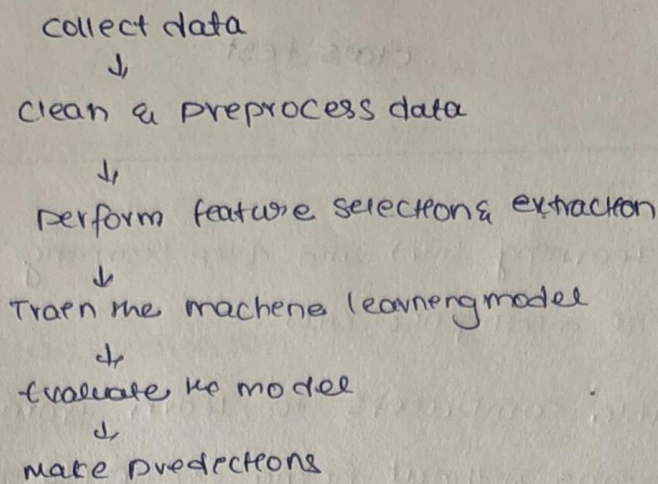
Introduction:

- machine learning (ML) and deep learning are two important branches of Artificial intelligence (AI)
- Both enable computers to learn from data & make predictions or decisions without being explicitly programmed.
- machine learning relies on algorithms that require manual feature engineering, whereas deep learning uses multi-layered neural networks to automatically learn complex features from raw data.

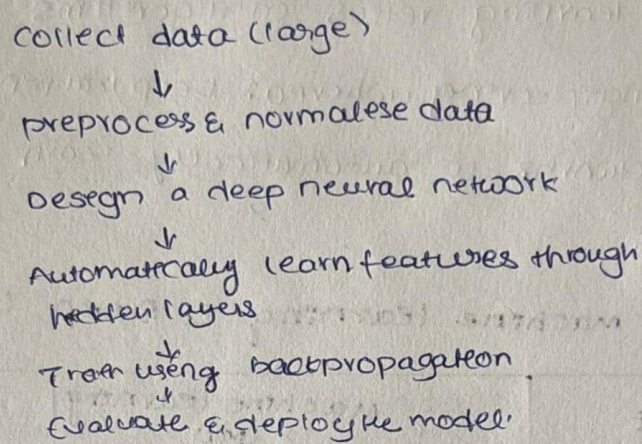
Diff b/w machine learning & Deep learning.

Aspect	machine learning (ML)	Deep learning (DL)
Architecture	uses algorithms such as Decision Trees, Random forest, SVM, KNN.	uses Artificial Neural Networks (ANN), CNN, RNN, LSTM & Transformers
Data requirement	performs well with small or medium sized datasets	Requires very large data sets to achieve high accuracy
Feature extraction	features must be manually selected & engineering by experts	features are automatically learned from raw input data
Training time	faster because models are simpler	slower because deep networks contain millions of parameters
Hardware Requirement	usually runs efficiently on CPUs	often requires GPUs or TPUS for efficient training.

machine learning workflow



Deep learning workflow



Advantages:-

machine learning:-

- Requires less Computational power
- Faster model training
- Easy to interpret
- Suitable for structured datasets.

Deep learning:-

- Automatically learns features
- excellent performance on complex problems
- High accuracy with large datasets
- Suitable for image, speech & natural language

Applications

ML:-

- spam email detection
- credit card fraud detection

DL:-

- facial recognition
- Autonomous vehicles

②

Introduction:

Representation learning is deep learning technique in which the model automatically learns meaningful features from raw input data instead of relying on manually designed features.

Advantages:

- Eliminates manual feature extraction
- Improves prediction accuracy
- Learns hierarchical features
- Work effectively with images, speech and text

Effect of width of a Neural Network

Width refers to the number of neurons in each hidden layer

Impact:

- Learns more features in a single layer
- Increases model capacity
- Requires more memory and computation
- Too much width may cause overfitting

Effect of Depth of a Neural Network:

Depth refers to the numbers of hidden layers.

Impact:

- Learns hierarchical representations
- Early layers learn simple features
- Deeper layers learn complex patterns
- Improves accuracy for complex tasks
- Excessive depth may lead to vanishing gradients and longer training time.

Comparison

Factor	wide Network	Deep Network
No. of Neurons	more	moderate
Hidden layers	few	many
Feature learning	Limited	Hierarchical
Computational cost	High memory	High computation
Best for	Simple problems	Complex problems

Applications:

- Image recognition
- Speech Recognition
- medical diagnoses
- Recommendation system
- NLP
- Autonomous vehicles

Conclusion:

Representation learning allows neural networks to automatically learn useful features from raw data. width increases the learning capacity by adding more neurons, whereas depth enables hierarchical feature learning. A proper balance between width & depth leads to better performance & Generalization.

③

feature	ReLU	LReLU	ELU
formula	$f(x) = \max(0, x)$	$f(x) = x, x > 0;$ $\alpha x, x \leq 0$	$f(x) = x, x > 0; x \leq 0;$ $\alpha(e^x - 1)$
+ve values	linear	linear	linear
-ve values	zero	small -ve	smooth exponential value
vanishing gradient	reduced	reduced	reduced
Dead Neuron Problem	can occur	mostly avoided	avoided
computation	fast	fast	slightly slower

ReLU

Advantages:

- simple & fast
- faster convergence
- Reduces vanishing gradient

Disadvantages:

- Dead neuron problem
- outputs zero for -ve inputs

Use cases:

- CNNs
- Image classification

LReLU

Advantages

- prevents dead neuron
- small gradient for -ve inputs
- Better than ReLU in many cases.

Disadvantages:

- Requires choosing the leakage parameter (α)

Use Cases:

- Deep CNNs
- Object Detection

ELU

Advantages:

- Smooth activation
- Faster convergence
- Better learning for deep networks

Disadvantages:

- Higher computational cost
- More complex than ReLU

Use Cases:

- Very Deep Neural Networks
- Medical Image Analysis

Conclusion:-

ReLU is the simplest & fastest activation function, leaky ReLU reduces the dead neuron problem & ELU provides smoother learning & often better performance in deep neural networks.

(u)

Introduction:

Unsupervised learning trains a neural network using unlabeled data to discover hidden patterns, structures or representations without predefined target output.

Applications:

- Clustering
- Anomaly detection
- Dimensionality Reduction
- Recommendation systems.

Restricted Boltzmann machine (RBM)

working principle:

1. Input data is given to the visible layer
2. Hidden layer learns latent features
3. Connections exist only between visible & hidden layers
4. Network reconstructs the input data
5. Weights are updated to minimize reconstruction error

Applications:

- movie recommendation
- Collaborative filtering
- Feature learning
- Dimensionality reduction

Autoencoders:

working principle:

1. Input data is fed into the encoder
2. Encoder compresses data into a latent representation
3. Decoder reconstructs the original input
4. Reconstruction error is minimized during training
5. The encoder learns useful representations.

Applications:

- Image compression
- Anomaly Detection
- Noise removal
- Feature Extraction

Comparison

Aspect	RBM	Autoencoder
Learning type	unsupervised	unsupervised
Architecture	Visible & hidden layers	Encoder & Decoder
Objective	Learn probability distribution	Reconstruct input data
Feature learning	Latent features	Compressed representation
Training	Contrastive Divergence	Backpropagation
Applications	Recommendation systems	Dimensionality Reduction